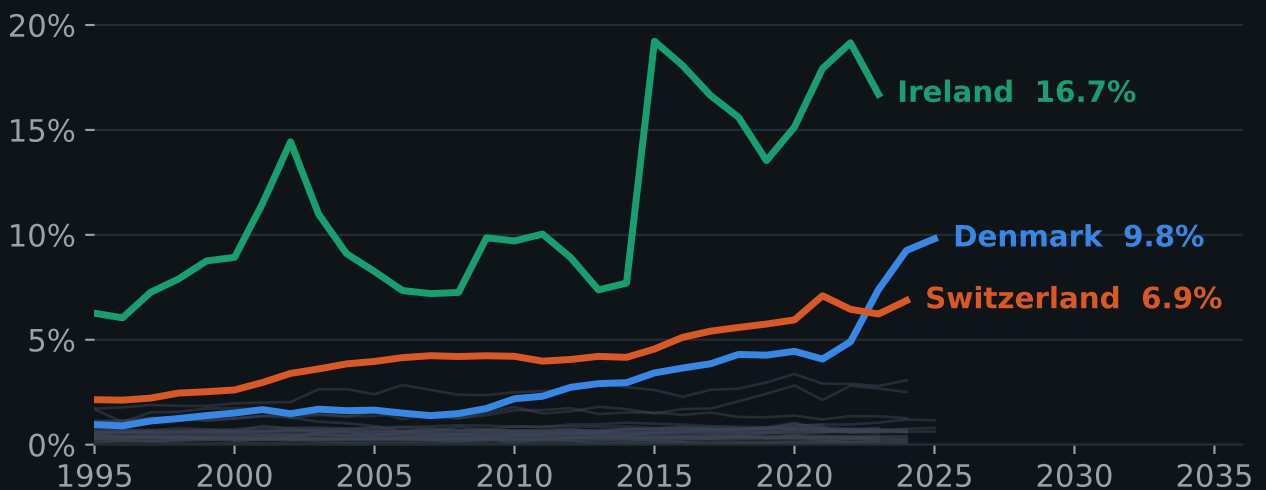


Pharmaceutical manufacturing in the national economy

42 countries · 35 years · how large the industry is, how concentrated it has become, and what each country makes.



Pharmaceutical share of total value added, 1995-2025.

Grey: every other country in the study.

0.5%

median country

16.7%

Ireland, the maximum

97%

Korean exports
that are biologics

Three findings

01

Scale is remarkably uniform.

In every large economy, pharmaceutical manufacturing accounts for approximately 1% of output — and has done so for three decades.

02

Concentration is the real variable.

Shares of 7–17% occur only in small economies where a single industry has come to dominate: Ireland, Denmark and Switzerland.

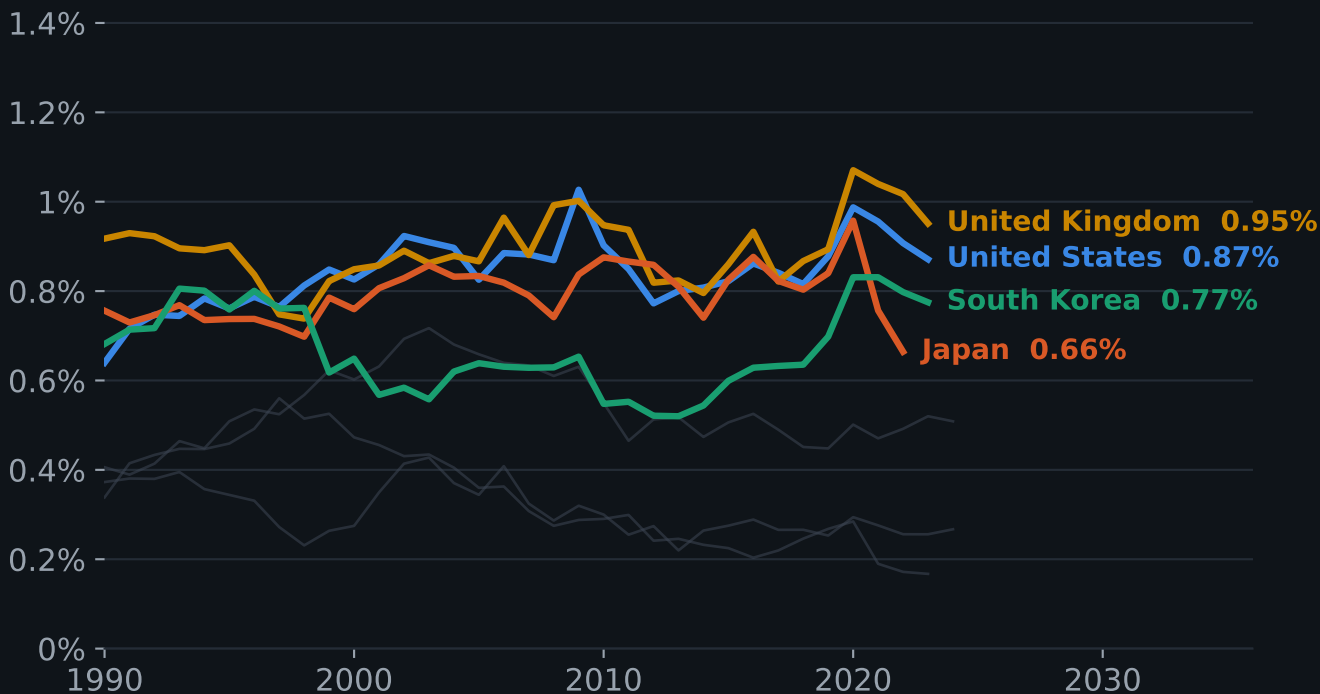
03

Composition differs more than scale.

Trade data separates biologics producers from finished-dose manufacturers — a distinction invisible in the headline economic figures.

Large economies hold pharmaceuticals near 1%

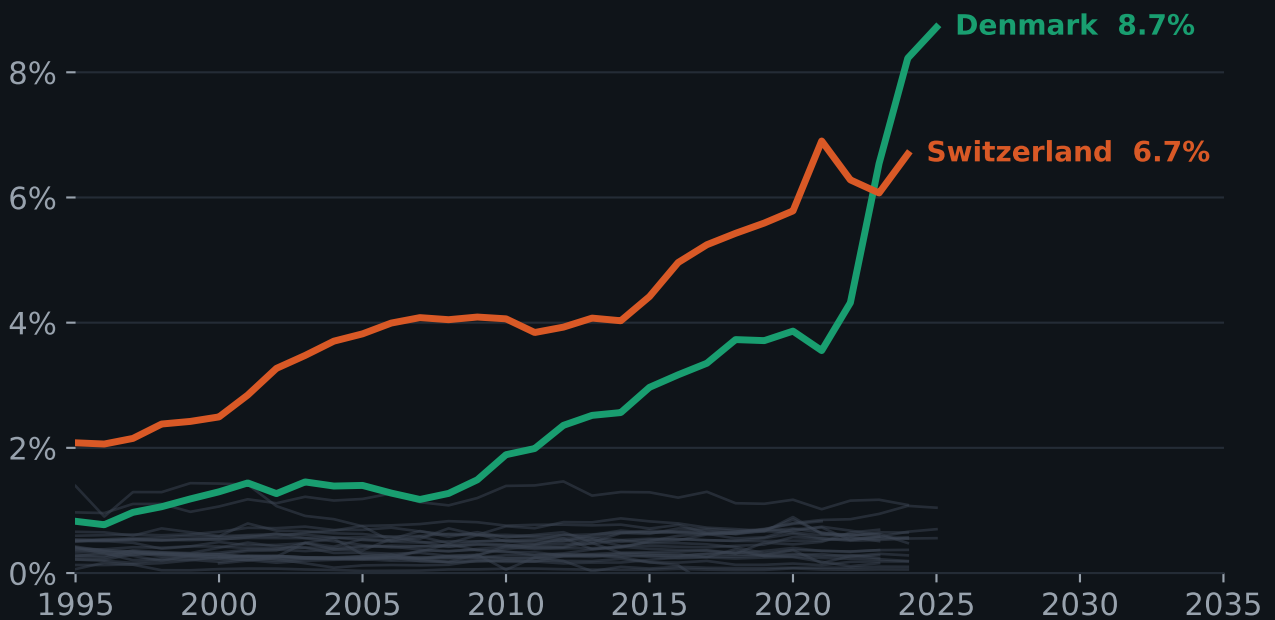
Share of total value added, 1990 to most recent year.



The band has held through the genomics era, the shift to biologics and the pandemic. Grey lines: Canada, Mexico, Australia, France, Italy, Spain, Germany.

The exceptions are all small economies

Pharmaceutical share of GDP, 1995–2025.



In 2023 Germany and Denmark each produced roughly €25bn of pharmaceutical value added. That is 0.6% of the German economy and 6.5% of the Danish one. Concentration, not sector size, determines macroeconomic exposure.

Two routes to concentration

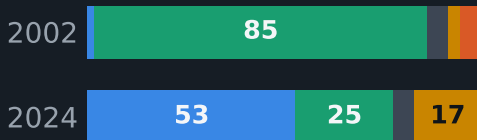
Ireland repositioned its output. Denmark deepened a single franchise. Both reached scale; the composition differs.

Ireland

6.3% → 16.7%

of value added, 1995–2023

Exports €15bn → €99bn



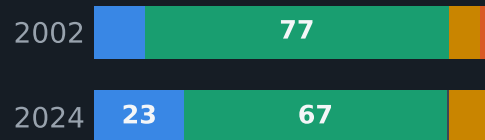
Export composition, %

Denmark

1.0% → 9.8%

of value added, 1995–2025

Exports €4bn → €22bn



Export composition, %

■ Biologics / vaccines

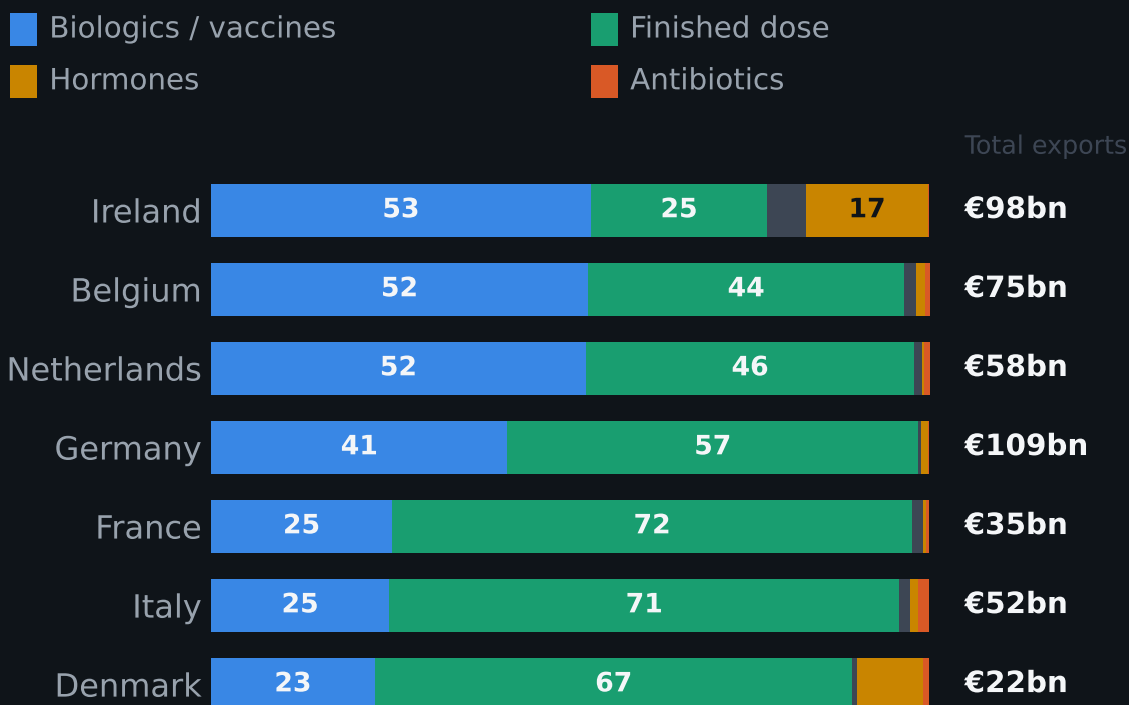
■ Finished dose

■ Hormones

Ireland moved from 85% finished dose in 2002 to 53% biologics and 17% hormones in 2024 — a change of product, not only of volume. Denmark stayed dose-led, scaling one franchise. Irish shares step up in 2015 with the national-accounts restatement; the export mix is unaffected by it.

Two industries share one label

Share of pharmaceutical exports by product class, 2024.



The same distinction holds outside Europe: 97% of South Korea's pharmaceutical exports to the EU are biologics, against 20% in 2013 — a national capability built within a decade.

What the analysis supports

Economic analysis

Danish, Irish and Swiss aggregates warrant an ex-pharmaceutical view before macro conclusions are drawn.

Industrial strategy

Composition can be changed within a decade — Ireland and South Korea repositioned output rather than expanding it.

Türkiye

0.35% of GDP, broadly unchanged since 2003 against a 0.5% median. The constraint is positioning, not data.

Method · NACE/ISIC C21 value added, Eurostat and OECD STAN, in national currency. Composition from HS chapter-30 trade, a proxy for output. Full sources and caveats online.

namikakmandev.github.io/pharma-gdp-share.html